

Exploratory Data Analysis (EDA): Two Variables

Jericho Lawson

MATH 240

Lecture #7 — Ch. 4.6, 5.1, 5.8

- Homework #2 is due **Friday** at 11:59pm
 - ▶ Includes textbook and R problems
 - ▶ Use of `dplyr`, `ggplot2`
- Checkpoint #2 on Friday
 - ▶ Work in groups of 2-3 on EDA tasks using R
 - ▶ 2 sets of problems, each related to a dataset
 - ★ 1 set will be graded for accuracy, the other for completion
 - ▶ Will have the whole period to work on them

Learning Objectives

- Analyze metrics of two or more variables using correlation
- Visualize multiple variables using faceted plots, box plots, and scatter plots using R

The Datasets

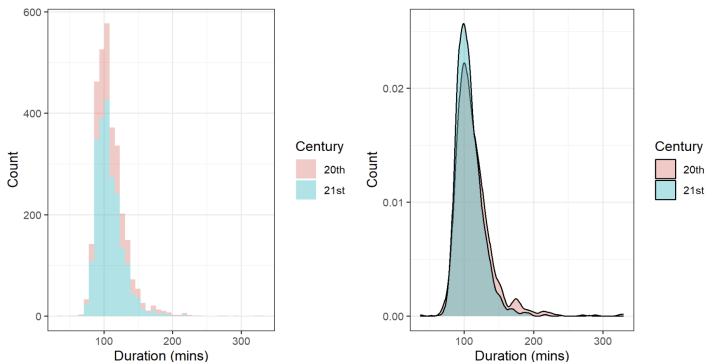
We will work with both the `movies` dataset from our end and the `duke_forest` dataset from the `openintro` package. We will investigate:

- Faceted plots
- Side-by-side box plots
- Scatter plots

Questions to consider:

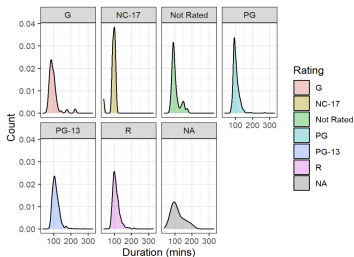
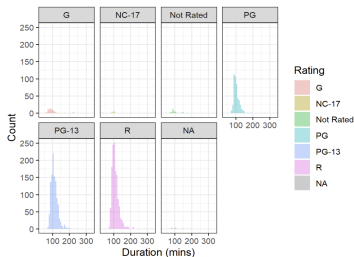
- How do we compare numerical variables across groups?
- How do we look at the association between two variables?

One Categorical and One Number Variable: Overlaid Density Plots



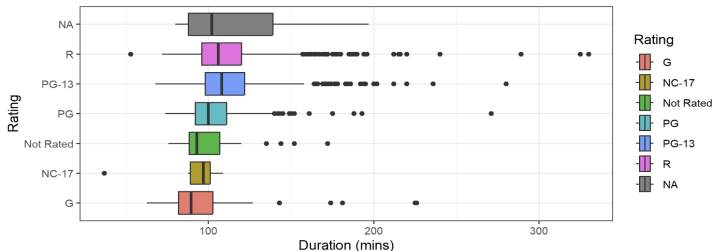
- Like looking at a single distribution: we compare the center, spread, and shape between the layers

One Categorical and One Number Variable: Faceted Histograms



- Useful when we have several groups (as well as multiple categorical variables)

One Categorical and One Number Variable: Side-by-side Box Plots



- Easy to compare metrics of center and spread directly
 - ▶ Overlaid histograms tell us more about the shape and density

One Categorical and One Number Variable: Five-Number Summaries

\$G

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
63.00	82.00	89.50	97.73	102.75	226.00

\$`NC-17`

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
37.00	89.00	97.00	93.23	101.00	109.00

\$`Not Rated`

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
76.0	88.5	93.0	103.1	107.0	172.0

\$PG

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
74.0	92.0	100.0	103.7	111.0	271.0

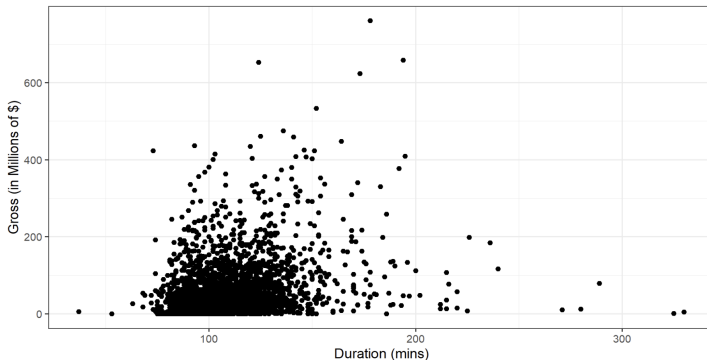
\$`PG-13`

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
68.0	98.0	108.0	111.8	122.0	280.0

\$R

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
53.0	96.0	106.0	110.4	120.0	330.0

Two Numerical Variables: Scatter Plots



- We care about the:
 - ▶ Direction: positive (upward) or negative (downward)
 - ▶ Strength: do the points surround a hypothetical line with little variability?
 - ▶ Linearity: is the trend linear?

Two Numerical Variables: Correlation

Correlation is the main metric we can use to identify how related one continuous variable is to another.

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) = \frac{s_{xy}}{s_x s_y}$$

- x_i is the i th data point for variable x
- y_i is the i th data point for variable y
- $\bar{x}$ and $\bar{y}$ are the means of x and y respectively
 - ▶ $\text{mean}(x)$, $\text{mean}(y)$
- s_x and s_y are the standard deviations of x and y respectively
 - ▶ $\text{sd}(x)$, $\text{sd}(y)$
- s_{xy} is the covariance of x and y
 - ▶ $\text{cov}(x, y)$
- n is the number of observations

Two Numerical Variables: Correlation

Correlation is the main metric we can use to identify how related one continuous variable is to another.

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) = \frac{s_{xy}}{s_x s_y}$$

- We care about the:
 - ▶ Direction: positive (+) or negative (-)
 - ▶ Strength: strong (closer to 1/-1), moderately strong (closer to 0.5/-0.5), weak (closer to 0), no strength (0)
 - ▶ Linearity: cannot infer from correlation

Activity: Two Variables

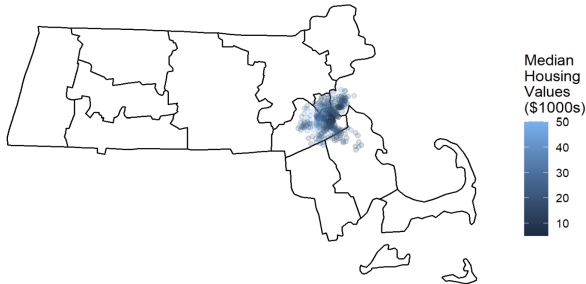
Using the `duke_forest` dataset, do the following in a pair or group of 3:

- Create an overlaid density plot using a numerical and categorical variable of your choice. Describe what you see in this plot.
- Create a side-by-side box plot using a numerical and categorical variable of your choice. Describe what you see in this plot.
- Create a scatter plot of two numerical variables of your choice. Describe the relationship between these variables and find the correlation using the code mentioned before.

Turn in a knitted `.pdf` file from a Quarto document that has all of this stuff in here under “Activity: 9/16” under Gradescope. This will not count as an assignment grade, but it will allow me to provide feedback on how you’re creating the plots and what you’re analyzing.

Aside: Mapping Data

Median Value of Owner-Occupied Homes
across Boston Census Tracts, 1970



- Can be seen as point or choropleth maps
- Example: NYC Trees

For Friday!

- Look through the R code discussed today!
- Checkpoint on Friday!
- HW due on Friday!